

Batch emissions report

exp2_rag_2026-09-07_14-25-34

Models: Unknown

Pipeline: RAG | Questions: 2 | Successful: 2 | Failed/unknown: 0

Batch runtime: 151.86 s | Successful requests/min: 0.79019

Measured cost (all requests)	Observed sum	Valid / all
Energy (Wh)	1.0232	2 / 2
Emissions (g CO2e)	0.72997	2 / 2

Method and data quality

Statistics and distributions use successful requests with finite measurements. Cost totals include measured failed requests. Missing/non-finite values are excluded, never replaced with zero. Partial totals are not estimates of unmeasured cost.

Variance and standard deviation use the sample definition ($n - 1$); both are N/A for fewer than two observations. Variance has squared units. P95 uses linear interpolation. Small values use significant digits or scientific notation.

Historical pipelines may already have stored missing telemetry as zero. This report cannot distinguish those defaults from measured zero. RAM fields describe estimated energy, not memory usage. Per-question charts use CSV row order; red points mark failed/unknown requests.

Metric	Missing	Zero
Latency (s)	0	0
Energy (Wh)	0	0
Emissions (g CO2e)	0	0
Retrieval latency (s)	0	0
Retrieval energy (Wh)	0	0
Retrieval emissions (g CO2e)	0	0
Generation latency (s)	0	0
Generation energy (Wh)	0	0
Generation emissions (g CO2e)	0	0
CPU energy (Wh)	0	0
GPU energy (Wh)	0	2
RAM energy (Wh)	0	0

Measurement scope and reliability

UNVERIFIED: This batch has no runtime measurement metadata. Its historical hardware coverage and backend methods cannot be established. Earlier zero defaults cannot be recovered.

A partial observed sum excludes unavailable readings. A complete batch cost cannot be claimed when coverage is incomplete. Zero readings are retained; unknown hardware coverage must not be interpreted as zero consumption.

Statistical summary

Latency (s)

Valid n	Mean	Median	Variance	Std dev	Min	Max	P95
2	72.927	72.927	25.856	5.0849	69.332	76.523	76.163

Energy (Wh)

Valid n	Mean	Median	Variance	Std dev	Min	Max	P95
2	0.51158	0.51158	0.0012085	0.034763	0.487	0.53616	0.53371

Emissions (g CO2e)

Valid n	Mean	Median	Variance	Std dev	Min	Max	P95
2	0.36498	0.36498	0.0006151	0.024801	0.34745	0.38252	0.38077

Retrieval latency (s)

Valid n	Mean	Median	Variance	Std dev	Min	Max	P95
2	0.24827	0.24827	0.050177	0.224	0.089875	0.40666	0.39082

Retrieval energy (Wh)

Valid n	Mean	Median	Variance	Std dev	Min	Max	P95
2	0.0015041	0.0015041	1.7424e-06	0.00132	0.00057069	0.0024375	0.0023441

Retrieval emissions (g CO2e)

Valid n	Mean	Median	Variance	Std dev	Min	Max	P95
2	0.0010731	0.0010731	8.8689e-07	0.00094175	0.00040715	0.001739	0.0016724

Generation latency (s)

Valid n	Mean	Median	Variance	Std dev	Min	Max	P95
2	72.679	72.679	23.628	4.8609	69.242	76.116	75.772

Generation energy (Wh)

Valid n	Mean	Median	Variance	Std dev	Min	Max	P95
2	0.51008	0.51008	0.0011184	0.033443	0.48643	0.53373	0.53136

Generation emissions (g CO2e)

Valid n	Mean	Median	Variance	Std dev	Min	Max	P95
2	0.36391	0.36391	0.00056927	0.023859	0.34704	0.38078	0.3791

CPU energy (Wh)

Valid n	Mean	Median	Variance	Std dev	Min	Max	P95
2	0.30915	0.30915	0.00042688	0.020661	0.29454	0.32376	0.3223

GPU energy (Wh)

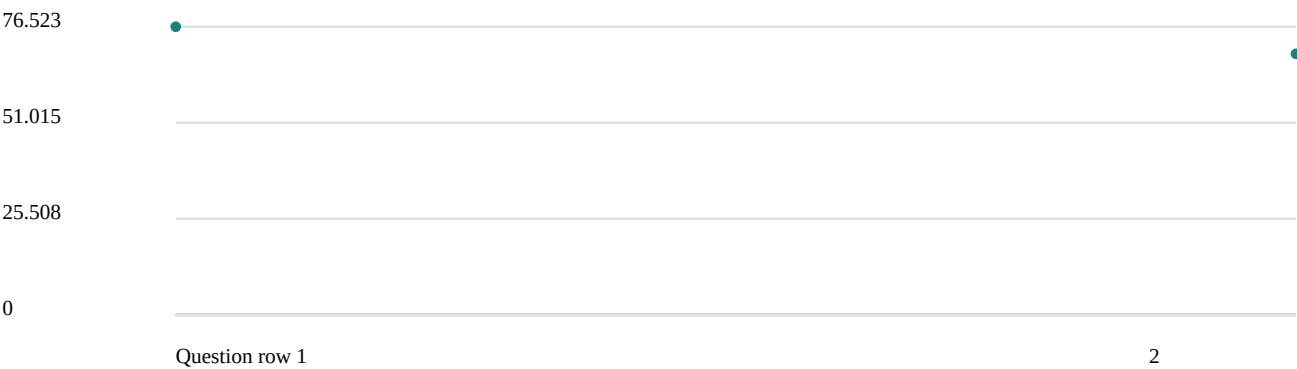
Valid n	Mean	Median	Variance	Std dev	Min	Max	P95
2	0	0	0	0	0	0	0

RAM energy (Wh)

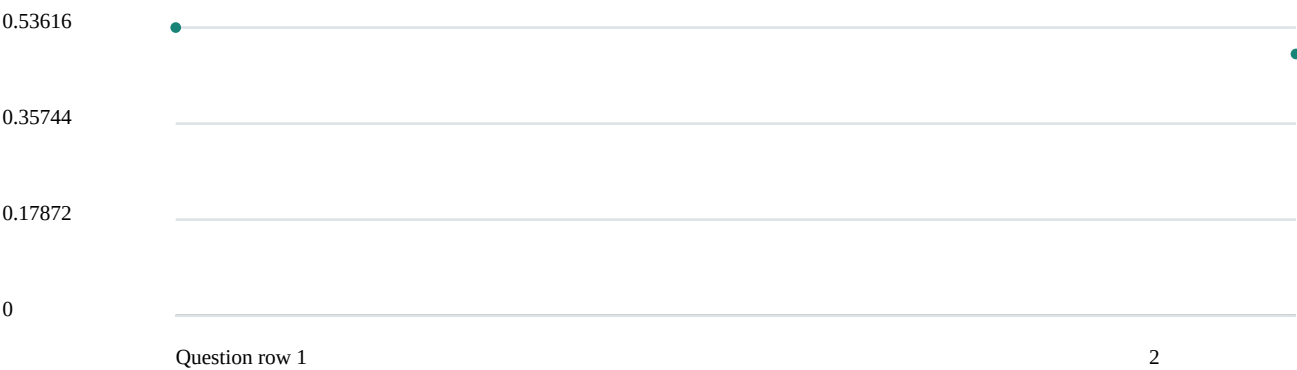
Valid n	Mean	Median	Variance	Std dev	Min	Max	P95
2	0.20243	0.20243	0.00019886	0.014102	0.19246	0.21241	0.21141

Per-question measurements

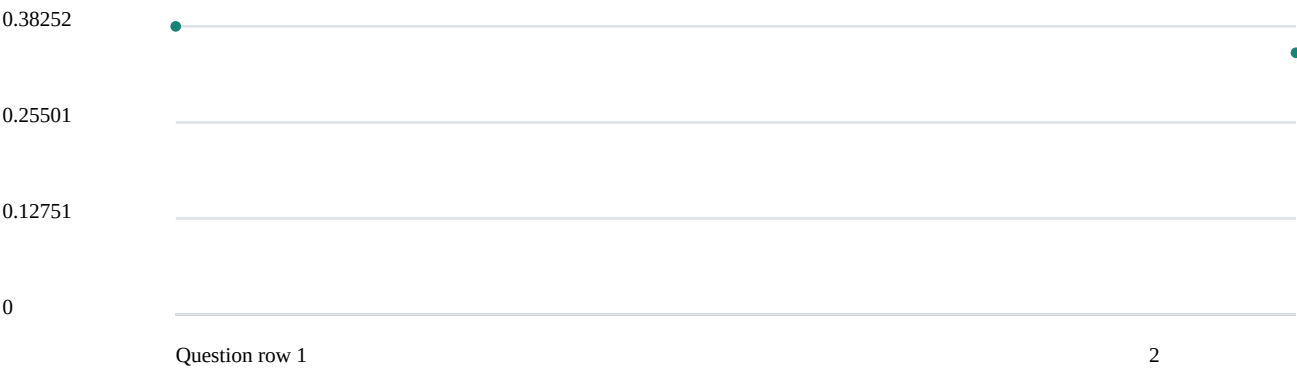
Latency (s)



Energy (Wh)



Emissions (g CO2e)



Distributions and energy breakdown

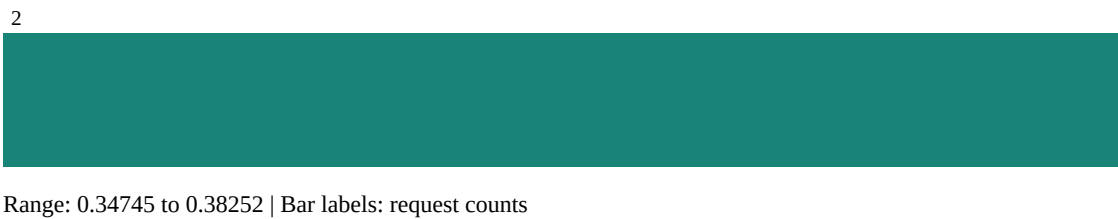
Latency (s)



Energy (Wh)



Emissions (g CO2e)



Hardware energy totals (all measured requests)

Component	Observed Wh
CPU energy (Wh)	0.6183
GPU energy (Wh)	0
RAM energy (Wh)	0.40487

RAG phase totals (all measured requests)

Phase metric	Observed sum
Retrieval latency (s)	0.49654
Retrieval energy (Wh)	0.0030082
Retrieval emissions (g CO2e)	0.0021461
Generation latency (s)	145.36
Generation energy (Wh)	1.0202
Generation emissions (g CO2e)	0.72782